

Station 1: Ore You Kidding

In the table below, fill in the corresponding blanks with the correct mineral, common name, and/or associated ore. (8)

Specimen	Common Name	Ore of...
1.	Brown Iron Ore	Iron
2.	TV Rock/TV Stone	3.
Olivine	4.	5.
Chalcedony	6.	Silicon
7.	Mock Ore, Rubyblende	8.

Identify the minerals from the following descriptions: (4)

9. Found in abundance in the Michigan Basin, is an evaporite, sometimes occurs as hopper crystals, soluble in water.

10. New Hampshire's state mineral, is an ore of tin and tungsten, well known varieties include emerald and aquamarine.

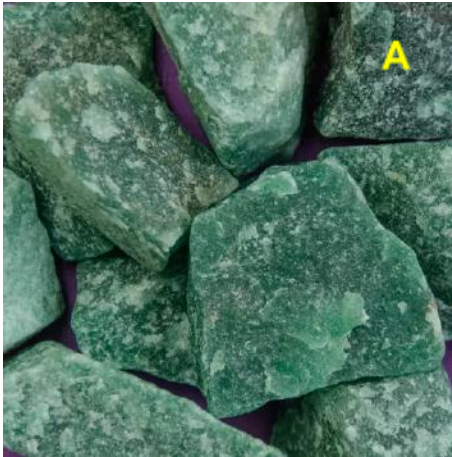
11. Named after the Ancient Greek word for moon, has a Mohs hardness of 2, fibrous varieties exhibit chatoyancy. (TB1)

12. Component of metamorphic schists, called "books" when found in large chunks, usually black, massive to platy crystal habit.

Station 2: Quartz the Matter?

Identify Specimens A-D and be specific. (Don't just say quartz) (4)

13.



14.



15.



16.



17. In which environment would you **most expect** Specimen B?

- a. Soils
- b. Volcanic rock nodules
- c. Beaches
- d. Alluvial Fans

18. Specimen C and D are most commonly observed with what type of twinning?

- a. Contact Twinning
- b. Penetration Twinning
- c. Cyclical Twinning
- d. Macle Twinning

19. Specimen A often has _____ inclusions, giving it a silvery green or blue sheen.

- a. Magnetite
- b. Albite
- c. Fuchsite
- d. Silver

20. Specimen D is known for its many health benefits including:

- a. Lowering blood pressure
- b. Beneficial to the immune system
- c. Relieves migraine symptoms
- d. Lowers cholesterol

21. What is the chemical composition of Pyrite?

Station 3: It's a Habit

Pick the letter from the answer bank that best describes these pictures. (5)

- A. Druzy
- B. Cubic
- C. Fibrous
- D. Tabular

- E. Stellate
- F. Rosette
- G. Acicular
- H. Oolitic

22.



24.



25.



23.



26.

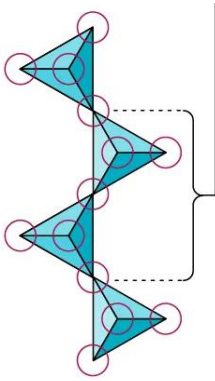


27. The texture of an igneous rock containing large crystals in a matrix of fine ground mass is considered to be _____. The included crystals are referred to as _____. (2)

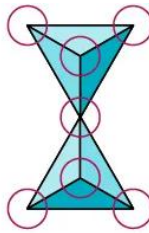
28. Kaolinite can contain what radioactive elements? (2)

Identify the following silicate classifications.

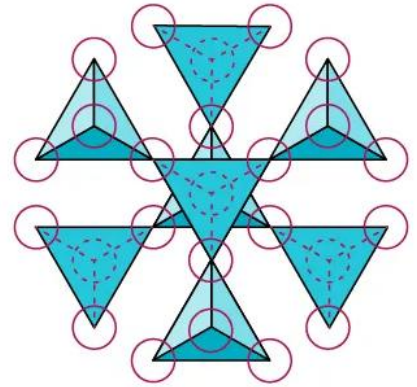
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30.



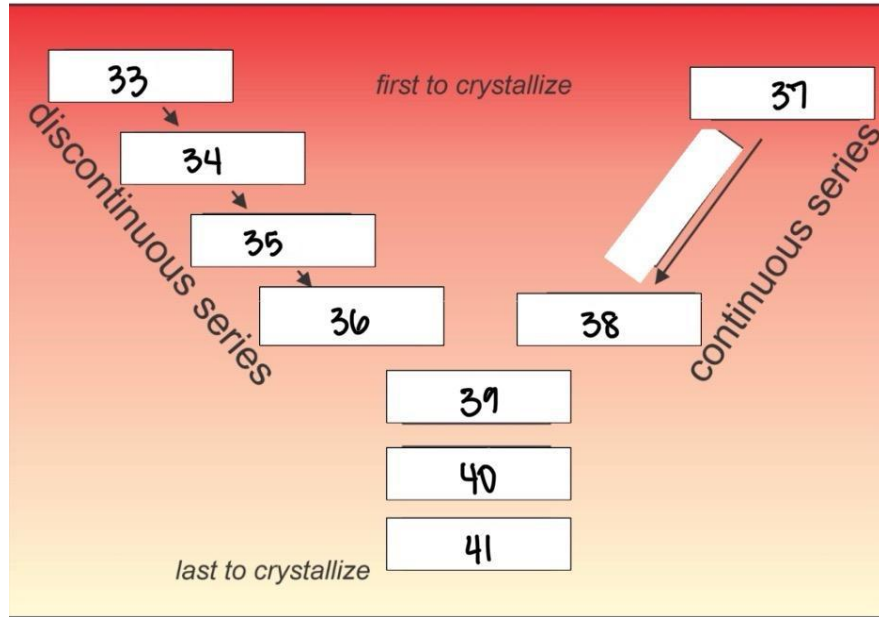
31.



32. Describe pleochrism and name one mineral on the Rocks and Minerals 2026 List that displays this. (3) (TB2)

Station 4: Norman Bowen

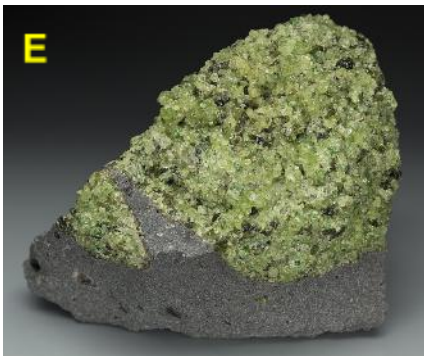
On your answer sheet, fill out the diagram. Write the corresponding specimen with its number. (4.5)



42. At higher temperatures, crystals form in more _____ magma, while at lower temperatures, crystals form in more _____ magma. (2)

Identify specimens E-G.

43.



44.



45.



46. For each specimen, identify whether the environment of formation is intrusive or extrusive. (4)

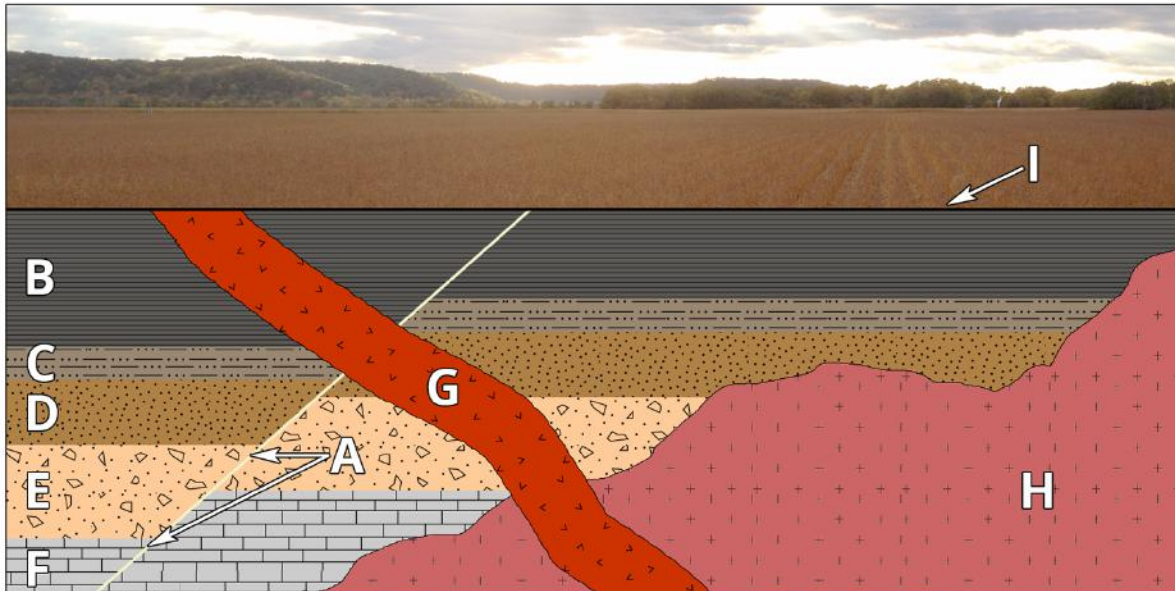
47. Explain the difference between intrusive and extrusive formation. (2) (TB3)

48. Which of the following is NOT an element found in the composition of Specimen F?

- a. Aluminum
- b. Magnesium
- c. Potassium
- d. Oxygen

Station 5: Shapes and Sizes

For questions 49-51, use the following diagram.



49. In this figure, which is the youngest?

- a. Fault A
- b. Intrusion G
- c. Rock Layer D

50. Your answer to question 74 demonstrates understanding of which relative dating principle?

51. Using the diagram above, order the letters from oldest to youngest. (4.5)

52. Consider the Grand Canyon of Arizona.

- i. Near the bottom, horizontal strata of the valley overlie tilted strata. This is an example of a/an _____.
- ii. The rock layers are continuous, found on both sides of the Grand Canyon, which has been carved by the Colorado River. They exhibit which principle?

53. What is a miscibility gap? (3)

What sedimentary structures are seen here?

54.



55.



56.



57.



58. Arizona is considered the _____ state due to the abundance of this mineral resource. (TB4)

59. The removal of crystals during cooling to filter out specific crystals and isolate them is known as:

- a. Grain Sorting
- b. Flux Melting
- c. Fractional Crystallization
- d. Fragment Crystallization

Station 6: The Same... or Different?



For questions 60-66, answer with the letter above that matches the description.

- 60. Considered a hopper crystal.
 - 61. Has an optical property known as double refraction or birefringence.
 - 62. Often associated with petroleum deposits.
 - 63. Impurities in this specimen can fluoresce yellow or green
 - 64. Fluoresces bright orange-red.
 - 65. Soluble with effervescence.
 - 66. Its high boron content can be used for many branches of industry including fiberglass, soaps, fire retardant, and pottery glaze.
67. There are many pseudomorphs that exist on your 2026 Rocks and Minerals list. Describe pseudomorphs and name one pair found in the Carbonate Family. (2)
68. The pair described in question 67 forms from the weathering of what elemental ore?

Match the following minerals to the correct formation description. (3)

- 69. Corundum
- 70. Limonite
- 71. Fluorapatite

- a. Hydration of hematite and magnetite, and chemical weathering of other iron rich minerals such as olivine and pyroxene
- b. Often formed through metamorphism of marbles, mica schist, and gneiss.
- c. Commonly occurs as a diagenetic mineral in sedimentary rocks.

Station 7: Magnetic and Expressive

Match the mineral to the correct magnetism it displays.

72. Chalcopyrite	a. Weakly magnetic
73. Almandine	b. Magnetic when heated
74. Tourmaline	c. Variably magnetic
75. Garnet Group	d. Antiferromagnetic
76. Goethite	e. Very magnetic with high iron content

For questions 77-83: Identify the mineral from the following descriptions.

77. Royal blue tectosilicate, specific gravity of 2.27-2.33, white streak, fluorescent variety known as “Yooperlite.”

78. Component of metamorphic schists, called “books” when found in large chunks, usually black, massive to platy crystal habit.

79. Derives its name from a famous river, green tectosilicate, weak olive-green UV fluorescence, streaks white.

80. Iridescent feldspar mineral, white streak, finest variety known as spectrolite, exhibits property known as Schiller Effect.

81. Fluorescent with crystal twinning, Moh’s hardness 4, exhibits thermoluminescence.

82. In the field, you find what *looks* like Olivine embedded in quartz. As a rocks and minerals expert, you correctly identify that this is **NOT** Olivine, but a different mineral entirely. What mineral is this? (1) (TB5)

Match the following to their parent rock: (4)

83. Gneiss

84. Marble

85. Slate

86. Quartzite

a. Sandstone

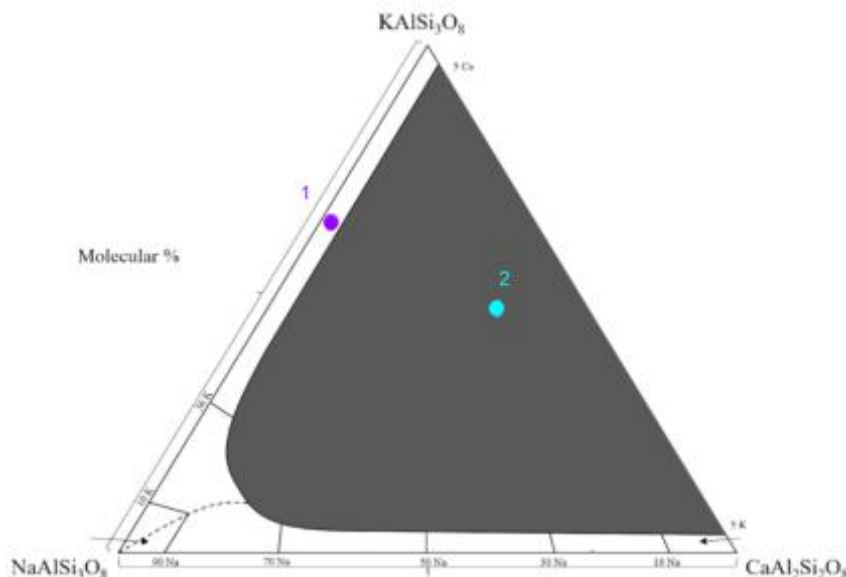
b. Shale

c. Diorite

d. Limestone

Station 8: Binary and Ternary

Consider the feldspar ternary diagram below.



87. Point 1 on the diagram corresponds to which type of feldspar, as appears on your official list?

88. If this diagram instead was a mica group solution series diagram, what would the miscibility gap represent?

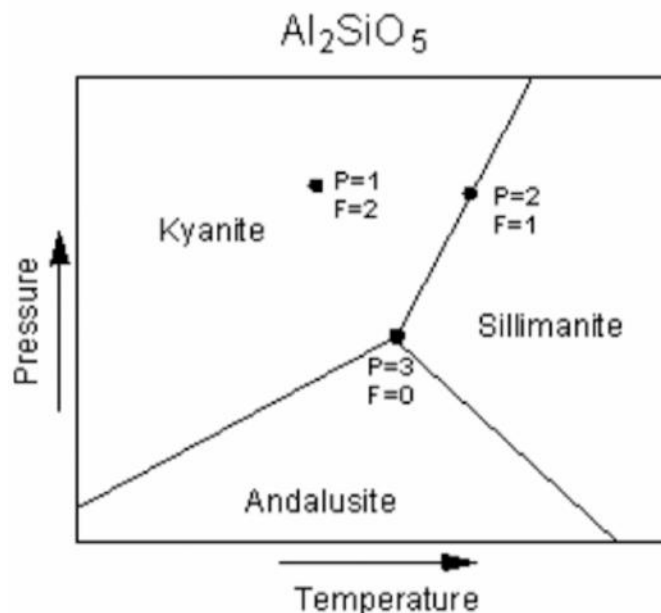
89. Which of these properties are influenced by the crystal structure of a mineral?

- A) Hardness
- B) Cleavage
- C) Diaphaneity
- D) A & B
- E) All of the above

90. What can you infer from the point where the three univariant lines intersect on the diagram to the right?

91. Agates are very commonly found along the north shores of Lake Superior. Can you shed some light on their origin?

92. Low-grade metamorphism is characterized by an abundance of _____ minerals.



Match the following specimens to their depositional environments. (5)

93. Chert

94. Lignite

95. Coquina

96. Conglomerate

97. Arkose

A) Swamp

B) Beach

C) Deep sea

D) Alluvial Fan

E) Glacial Till

Station 9: Rock Climbing... ish

Identify the specimens K-N.

98.



K

99.

L



100.



M

101.

N



For questions 102-106, write the letter from above that matches the description below.

102. Found as a secondary mineral in clay deposits and an aluminum ore (the world's main source of aluminum).

103. Found in hydrothermal veins and has the same crystal structure as sphalerite.

104. Has good fissility and means "leaf" in Greek.

105. Important ore of lead and silver and the official state mineral of Missouri.

106. Important ore of copper and is soluble in HNO_3 .

107. Order the metamorphic process of shale from lowest to highest grade metamorphism. (2.5)

108. What is the carbon content of Anthracite? (TB6)

109. Anthracite accounts for what percent of global coal reserves?

Station 10: Butterflies have Metamorphosis too

For questions 110-114, answer with the specific type of metamorphism that matches the description.

110. Mineral composition of surrounding rocks is changed due to intense heat of intruding magma.

111. Also known as dynamic metamorphism, sedimentary rocks are folded due to compression forces from mountain building, altering the original rock form.

112. Occurs as a result of mechanical deformation where heat is generated by the friction of two rock bodies along a shear zone.

113. Occurs when an extraterrestrial body, such as a meteorite, impacts with the Earth or if there is a very large volcanic explosion.

114. Minor alteration of chemical composition of rocks caused by the weight and pressure of water mass and chemically active hot gases/water vapor.

115. Along a typical geothermal gradient, which sequence correctly represents the metamorphic progression of shale with increasing temperature and pressure?

- a. Slate → Phyllite → Schist → Gneiss → Migmatite
- b. Slate → Hornfels → Eclogite → Gneiss → Migmatite
- c. Phyllite → Slate → Schist → Migmatite → Gneiss
- d. Zeolite → Blueschist → Greenschist → Hornfels

116. Which metamorphic facies are associated with low temperature and high pressure conditions?

- a. Greenschist
- b. Amphibolite
- c. Blueschist
- d. Hornfels

117. In a metamorphic facies diagram, what does the label “Conditions that don’t exist on Earth” imply?

- a. Rocks cannot be stable at those temperatures
- b. No rocks reach those P–T conditions naturally
- c. The conditions exist only in the mantle
- d. It’s a zone of constant metamorphic grade

Match the following. A letter may be used more than once. (5)

118. Citrine

119. Tourmaline

120. Chalcedony

121. Almandine

122. Rose Quartz

a. Hardness of 6.5-7

b. $\text{Fe}_3^{2+}\text{Al}_2\text{Si}_3\text{O}_{12}$

c. Vitreous to Resinous luster

Station 11: Astrology and Numerology (kinda)

Identify the Specimens O-R.

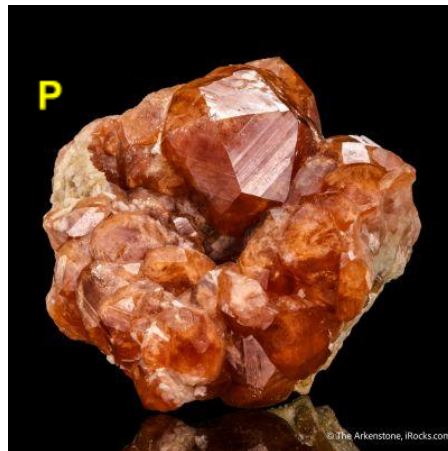
123.

O



124.

P



125.

Q



126.

R



For questions 127-130, answer with the correct letter from above.

127. Transparent versions of this mineral are used as gems and ornamental stones in India known as Shajar.

128. Once used to ward off witchcraft and disease and is considered an index mineral.

129. In Ancient Greece, it was considered to be a strong antidote against drunkenness.

130. Considered a potent prosperity stone that enhances relaxation and cooperation.

131. Which factor *most directly limits* the extent of solid solution between two minerals at lower temperatures?

- a. Ionic size difference
- b. Crystal field stabilization energy
- c. Entropy of mixing
- d. Decrease in diffusion rates

132. Which of the following sequences correctly represents the order of mineral precipitation during the evaporation of seawater? (i.e. evaporite sequence)

- a. Halite → Gypsum → Sylvite → Carbonates
- b. Carbonates → Gypsum → Halite → Potassium salts
- c. Gypsum → Carbonates → Halite → Sylvite
- d. Carbonates → Halite → Gypsum → Potassium salts

133. Which depositional environment is **least favorable** for large evaporite deposits?

- a. Alluvial Fan
- b. Closed marine basin
- c. Continental rift valley
- d. Deep open ocean

134. Although dolomite can form in association with evaporites, it is not a primary evaporite mineral. Rather, it is often formed by _____ replacement of limestones.

Station 12: ID Frenzy

Identify the following specimens. Be specific when you can (no “gypsum” or “quartz”).

126. (ID the blue mineral)



127.



128.



129.



130.



131.



132.



133.



134.



135.



136.



137.



138.



139.

